

## 1.3 Subgroup, Quotient Group

June 11, 2026

- Let  $G$  be a topological group and  $H \subset G$ . Then  $H$  is a subgroup of topological group  $G$  iff  $H$  is a subgroup of group  $G$  and  $H$  is a closed set in topological space  $G$ .

*Remark 0.1.*  $H$  is a topological group equipped with induced topology on  $G$ . A subgroup of an abstract group which is a topological group is a topological group. This works as restriction of continuous maps is continuous.

- A normal subgroup  $N$  of topological group  $G$  if  $N$  is a normal subgroup of abstract group  $G$ .

**Proposition 0.1.** Let  $G$  be a topological group. Let  $H$  be a subgroup of abstract group  $G$ .

- (i)  $\overline{H}$  is a subgroup of top. group  $G$
- (ii) If  $H$  is normal subgroup of  $G$  then  $\overline{H}$  is normal.

*Proof.* (i) Observe  $\overline{H} = \bigcap_{V \in \mathcal{V}} HV$  where  $\mathcal{V}$  is collection of all nbd of  $e$ . As  $H \subseteq \overline{H}$  hence  $\overline{H} \neq \emptyset$ .

For  $x, y \in \overline{H}$  we want to show  $xy^{-1} \in \overline{H}$ .

We know  $m : G \times G \rightarrow G$  is continuous.

$$(x, y) \mapsto xy^{-1}$$

Observe  $m^{-1}(\overline{H})$  is closed, and  $H \times H \subseteq m^{-1}(\overline{H})$

Hence  $\overline{H} \times \overline{H} = \overline{H \times H} \subseteq \overline{m^{-1}(\overline{H})}$  and this implies

$$m(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

□

Cii) We construct a mapping

$$n : G \times G \rightarrow G$$

$$(x, y) \mapsto xyx^{-1}$$

and observe  $n$  is a continuous map this implies  $n^{-1}(\overline{H})$  is closed and similar to (Ci)

$$H \times H \subseteq n^{-1}(\overline{H})$$

this implies  $\overline{H} \times \overline{H} = \overline{H \times H}$  is a subset of  $n^{-1}(\overline{H})$ . Hence

$$n(\overline{H} \times \overline{H}) \subseteq \overline{H}.$$

\* Set of all left cosets set  $G/H$ . We have a natural map or projection  $\pi : G \rightarrow G/H$ .

$$x \mapsto xH$$

*Remark 0.2.* If  $G$  is a topological group, we will topologize  $G/H$  assuming  $H$  is closed. Construct

$$\tau_{G/H} = \{O \subseteq G/H \mid \pi^{-1}(O) \in \tau_G\}$$

i.e.  $O$  is open if and only if  $\pi^{-1}(O)$  is open in  $G$

which ensures  $\pi$  is a continuous map, moreover a quotient map where  $\pi$  is surjective by construction.

**Lemma 0.1.**  $G/H$  is a  $T_1$ -space, and  $\pi$  is open.

*Proof.* Since  $H$  is closed then  $\alpha H$  is closed by homeomorphism. Observe  $G - \alpha H$  is open in  $G$ . Let  $\tilde{\alpha} = \alpha H \in G/H$ . Observe  $\pi^{-1}(G/H - \tilde{\alpha}) = G - \alpha H$ . Hence  $\tilde{\alpha}$  is closed,  $G/H$  is  $T_1$ .

We want to show  $\pi$  is an open map. Let  $O$  be an open map in  $G/H$  then  $\pi(O)$  is open  $G/H$  iff  $\pi^{-1}(\pi(O))$  is open in  $G$ . Now  $OH = \pi^{-1}(\pi(O))$  and  $OH = \bigcup Oh$ . Hence  $OH$  is open. So  $\square$

$\pi^{-1}(\pi(O))$  is open which implies  $\pi(O)$  is open.

**Lemma 0.2.**  $G/H$  is a  $T_2$  space.

*Proof.* Consider  $xH \neq yH$  in  $G/H$ .

Now  $x \in xH$  and  $x \in G \setminus yH$  which is open as  $H$  is closed and  $yH$  is closed. Now, let  $V$  be a neighborhood of  $x$  such that  $x \in V \subset G \setminus yH$ .

Observe  $Vx \cap yH = \emptyset$  implies  $VxH \cap yH = \emptyset$  otherwise  $vh_1 = gh_2$  and  $vx = yh_2h_1^{-1}$  implying  $Vx \cap yH \neq \emptyset$ . Now, we can find a neighborhood of  $V_1$  such that  $V_1^{-1}V_1 \subset V$  therefore  $V_1^{-1}V_1xH \cap yH = \emptyset$  this implies  $V_1xH \cap V_1yH = \emptyset$ , which is  $\pi(V_1x) \cap \pi(V_1y) = \emptyset$  and by previous lemma  $\pi(V_1x)$  and  $\pi(V_1y)$  are open sets and  $xH \in \pi(V_1x)$  and  $yH \in \pi(V_1y)$ . Hence  $\square$

$G \setminus H$  is  $T_2$

*Remark 0.3.* We call  $G \setminus H$  the quotient space where  $H$  is a closed subgroup of  $H$ , has equipped with quotient topology.

**Lemma 0.3.** If  $G$  is a topological group,  $H$  is normal subgroup, the quotient topology equipped on the quotient group  $G \setminus H$  is a topological group.

*Proof.* We want to show  $\chi_1 : G \setminus H \times G \setminus H \rightarrow G \setminus H$ ,

$$(x_1, y_1) \mapsto x_1 y_1^{-1}$$

is continuous.

And  $\chi : G \times G \rightarrow G$ ,

$$(x, y) \mapsto xy^{-1}$$

is continuous  $\square$

$$\begin{array}{ccc}
G \times G & \xrightarrow{\pi \times \pi} & G \setminus H \times G \setminus H \\
\downarrow \pi & & \downarrow \psi_1 \\
G & \xrightarrow{\psi} & G \setminus H
\end{array}$$

Observe  $\psi_1(\pi \times \pi) = \psi\pi$  and  $\pi, \psi$  are continuous. Hence  $\psi_1(\pi \times \pi)$  are continuous.

Let  $O_1$  be an open subset of  $G \setminus H$  then

$\psi_1(\pi \times \pi)^{-1}(O_1)$  is open. As  $\pi \times \pi$  is an open map, observe

$\pi \times \pi(\pi \times \pi)^{-1}\psi_1^{-1}(O_1)$  is open. Hence  $\psi_1^{-1}(O_1)$  is open.

□

**Lemma 0.4.** Let  $G$  be a topological group, and  $H$  be a subgroup.

(a) If  $G$  is compact then  $H$  and  $G \setminus H$  are both compact.

(b) If  $G$  is locally compact then  $H$  and  $G \setminus H$  have both locally compact.

*Proof.* (a)  $H$  is a closed subset of a compact set and  $G/H = \pi(G)$  where  $G$  is compact and  $\pi$  is continuous mapping. Hence a continuous map of compact set is compact.

(b) A closed subset of a locally compact set is locally compact.

Let  $S$  be locally compact,  $T = \overline{S}$  and  $T \subseteq S$ . For  $p \in T \cap S$ , there exists a neighborhood of  $p$  in  $S$ ,  $\overline{U_p}$  st.  $\overline{U_p}$  is compact. Observe for  $p \in T$  there exists an open set  $T \cap U_p$  in  $T$  such that  $\overline{T \cap U_p} = T \cap \overline{U_p}$  is compact as  $T \cap \overline{U_p}$  is closed subset of  $\overline{U_p}$  which is compact.

Hence  $H$  is locally compact.

Goal:  $G/H$  is locally compact.

□

For  $q_i \in G \setminus H$  and nbd of  $q_i$ . Let  $U = \pi^{-1}(U_i)$ , and let  $x \in G$  st.  $\pi(x) = q_i$ .

As  $G$  is locally cpt we get a nbd of  $x$   $\overline{O}$  st.  $\overline{O}$  is cpt and  $\overline{O} \subseteq U$  by proposition given below. Then we have  $\pi(\overline{O}) \subseteq \pi(U) = U_i$ . If  $O_i = \pi(O)$  a nbd of  $q_i$ . And  $\pi(\overline{O})$  is a continuous image of cpt set hence we get  $\pi(\overline{O})$  is cpt. Since  $G \setminus H$  is  $T_2$ -space  $\pi(\overline{O})$  is closed which implies

$O \subseteq \overline{O} \Rightarrow O_i \subseteq \pi(\overline{O})$  and  $\overline{O_i} \subseteq \pi(\overline{O}) = \pi(\overline{O})$  which is cpt, so is  $\overline{O_i}$ . Thus  $O_i$  is a nbd of  $q_i$  such that  $\overline{O_i}$  is cpt. It follows  $G \setminus H$  is locally cpt.

Corollary

If  $G$  is locally cpt then  $G \setminus H$  is normal or  $T_4$ .

**Proposition 0.2.** For any topological group  $G$ , let  $U$  be a neighborhood of  $e$ , there exists a neighborhood of  $e$   $V$  such that  $V \subseteq U$ .

*Proof.* By property 5) and 6) we know that for every neighborhood of  $e$   $U$  we have a symmetric neighborhood of  $e$   $V$  such that  $V^2 \subseteq U$ . Now  $x \in \overline{V} = \bigcap_{w \in V^e} w$  where  $V^e$  is set of all

neighborhoods of  $e$ , this implies  $x \in \overline{V} \subseteq V^2 \subseteq U$ .

□